

Continuity is an important property of functions relying on limits.

Def: A function f is continuous at a if

$$\lim_{x \rightarrow a} f(x) = a$$

If f is continuous for all numbers in its domain, we just say f is continuous.

Notice in the definition that we require 3 things

1. $f(a)$ is defined
2. $\lim_{x \rightarrow a} f(x)$ exists
3. $\lim_{x \rightarrow a} f(x) = f(a)$

if any of these fail, the function f is discontinuous

Ex: $f(x) = \frac{x^2 - x - 2}{x - 2}$ is discontinuous at $x = 2$

Since it is undefined,

- It is continuous everywhere else on its domain as by direct substitution $\lim_{x \rightarrow a} f(x) = f(a)$

Just like we can have limits from the left or right, we can have continuity from the left and right.

Def: f is continuous from the left if

$$\lim_{x \rightarrow a^-} f(x) = f(a)$$

and continuous from the right if

$$\lim_{x \rightarrow a^+} f(x) = f(a)$$

Def: A function is continuous on an interval I if f is continuous for all values of x in the interval I .

Ex: $f(x) = 1 - \sqrt{1-x^2}$

If $-1 < a < 1$, then

$$\begin{aligned}\lim_{x \rightarrow a} f(x) &= \lim_{x \rightarrow a} 1 - \sqrt{1-x^2} \\ &= \lim_{x \rightarrow a} 1 - \sqrt{\lim_{x \rightarrow a} (1-x^2)} \\ &= 1 - \sqrt{1-a^2} \\ &= f(a)\end{aligned}$$

We also have

$$\lim_{x \rightarrow 1^-} f(x) = 1 = f(1)$$

and

$$\lim_{x \rightarrow -1^+} f(x) = 1 = f(-1)$$

So f is continuous on $[-1, 1]$

Always evaluating limits is annoying, So we have some properties to help us.

Thm: If f and g are continuous at a , and c is a constant number, then the following are also continuous at a :

- $f + g$
- $f - g$
- $f \cdot g$
- f/g $g(a) \neq 0$
- cf

Specifically, we can use this to see that

Thm: For every number in their domain,

- Polynomials
 - Root functions
 - Rationals
 - Trig functions
- are continuous.

Ex: a) $x^{100} - 2x^{37} + 75$ is continuous everywhere
since it is a polynomial

b) $\frac{x^2 + 2x + 17}{x^2 - 1}$ is a rational, so its
continuous on its domain

$$D = (-\infty, -1) \cup (-1, 1) \cup (1, \infty)$$

is where it is continuous

c) $h(x) = \sqrt{x} + \frac{x+1}{x-1} - \frac{x+1}{x^2+1}$ is a sum of roots
and rationals, each is continuous on their domain

$\sqrt{x}$ has domain $[0, \infty)$

$\frac{x+1}{x-1}$ has domain $(-\infty, 1) \cup (1, \infty)$

$\frac{x+1}{x^2+1}$ has domain $(-\infty, \infty)$

So $h(x)$ is continuous on its domain

$$D = [0, 1) \cup (1, \infty)$$

Function Composition:

Thm: If f is continuous at b and $\lim_{x \rightarrow a} g(x) = b$,
then

$$\lim_{x \rightarrow a} f(g(x)) = f\left(\lim_{x \rightarrow a} g(x)\right) = f(b)$$

In terms of Continuity

Thm: If g is continuous at a and f is continuous at $g(a)$, then the composite function $(f \circ g)(x)$ is continuous at a .

Ex: Where is $\sin(x^2)$ continuous?

$\sin(x^2) = (f \circ g)(x)$ w/ $f(x) = \sin(x)$, $g(x) = x^2$
both f and g are continuous everywhere,
s, $\sin(x^2)$ is continuous everywhere

Ex:

$$F(x) = \frac{1}{\sqrt{x^2+7}-4}$$

$$F(x) = (f \circ g \circ h \circ k)(x)$$

$$f(x) = \frac{1}{x}, \quad g(x) = x-4, \quad h(x) = \sqrt{x}, \quad k(x) = x^2+7$$

Each continuous on its domain, so F is continuous on its domain.

$$D = (-\infty, -3) \cup (-3, 3) \cup (3, \infty)$$

Ex: For what value of c is

$$f(x) = \begin{cases} cx^2 + 2x & x < 2 \\ x^3 - cx & x \geq 2 \end{cases} \quad \text{continuous?}$$

$$f(2) = 2^3 - c2 = 8 - 2c$$

$$\lim_{x \rightarrow 2^+} f(x) = f(2) \quad \text{as } x^3 - cx \text{ is continuous}$$

$$\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} Cx^2 + 2x = 4C + 4$$

Since

$Cx^2 + 2x$ is
continuous

$$\text{we want } \lim_{x \rightarrow 2} f(x) = f(2)$$

So, we need

$$4C + 4 = 8 - 2C$$

$$6C = 4$$

$$C = \frac{2}{3}$$

Ex: Suppose f and g are continuous s.t. $g(2) = 6$
and

$$\lim_{x \rightarrow 2} [3f(x) + f(x)g(x)] = 36$$

What is $f(2)$? $\left. \right\}$

$$3f(2) + f(2)g(2) = 36$$

$$3f(2) + 6f(2) = 36$$

$$9f(2) = 36$$

$$f(2) = 4$$

Ex:

$$f(x) = \begin{cases} \frac{x^2-4}{x-2} & x < 2 \\ ax^2-bx+3 & 2 \leq x < 3 \\ 2x-a+b & 3 \leq x \end{cases}$$

$$f(2) = 4a - 2b + 3$$

$$f(3) = 6 - a + b$$

$$\lim_{x \rightarrow 2^+} f(x) = f(2)$$

$$\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} \frac{x^2-4}{x-2} = \lim_{x \rightarrow 2^-} \frac{(x-2)(x+2)}{x-2} = \lim_{x \rightarrow 2^-} x+2 = \underline{\underline{4}} = f(2)$$

$$\lim_{x \rightarrow 3^+} f(x) = f(3)$$

$$\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^-} ax^2 - bx + 3 = \underline{\underline{9a - 3b + 3}} = f(3)$$

$$\begin{cases} 4a - 2b + 3 = 4 \\ 9a - 3b + 3 = 6 - a + b \end{cases} \Rightarrow \begin{cases} 4a - 2b = 1 \quad \times 2 \\ 10a - 4b = 3 \end{cases}$$

$$\Rightarrow \begin{cases} 8a - 4b = 2 \\ -(10a - 4b = 3) \end{cases}$$

$$\Rightarrow -2a + 0b = -1 \Rightarrow \boxed{a = \frac{1}{2}}$$

$$\text{If } a = \frac{1}{2}, \quad 8\left(\frac{1}{2}\right) - 4b = 2$$

$$4 - 4b = 2$$

$$-4b = -2$$

$$b = \frac{1}{2}$$

So, f is continuous if $a = b = \frac{1}{2}$